

Automated Detection of Volcanic Unrest Using Machine Learning on RCM InSAR Data





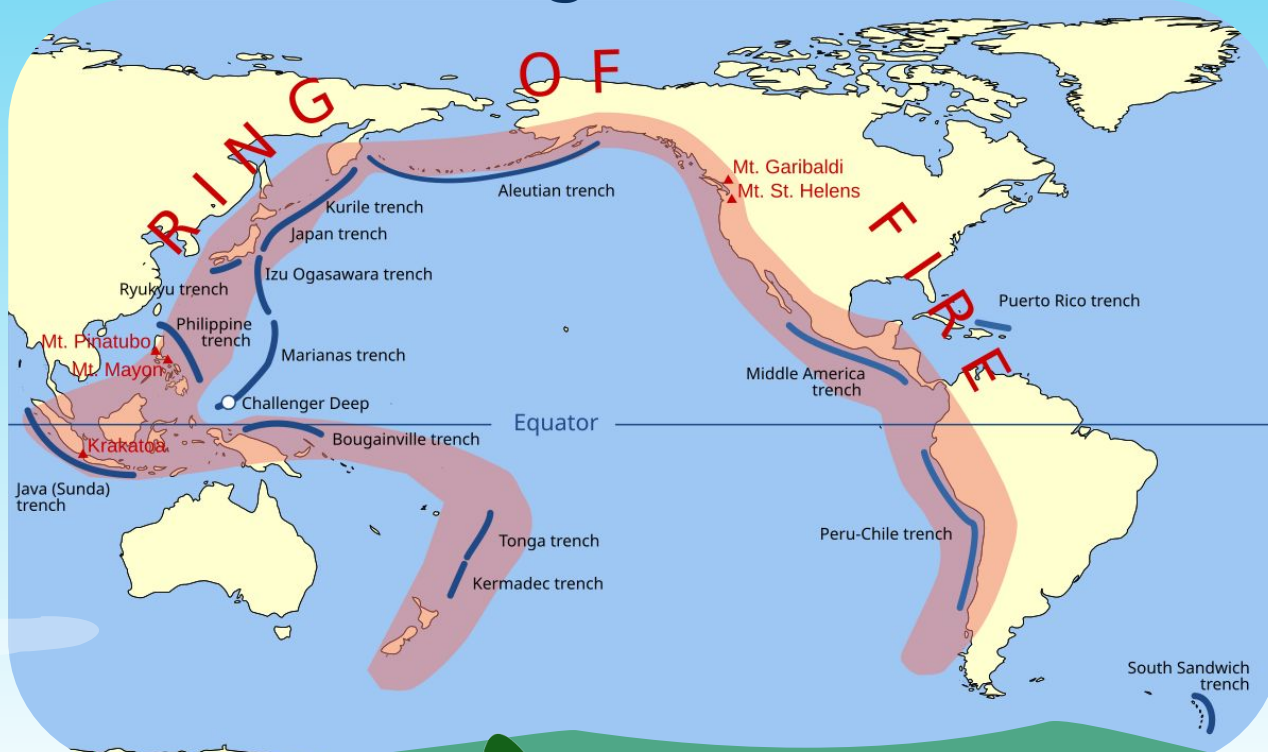
Kilauea (Hawaii)



Reykjanes (Iceland)



Ring of Fire



Specifically in Canada...

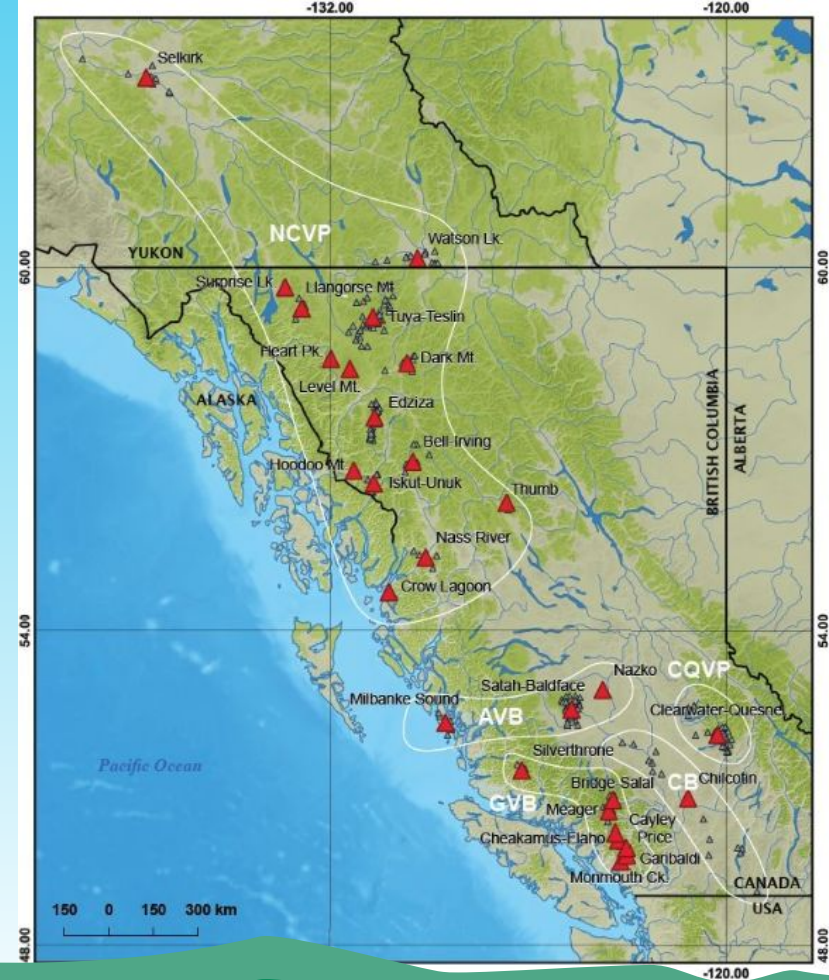
Volcanoes that have erupted within the past 1.8 million years



Volcano Groups with known hazards



Individual vents within these groups



Volcanic Eruption Impacts



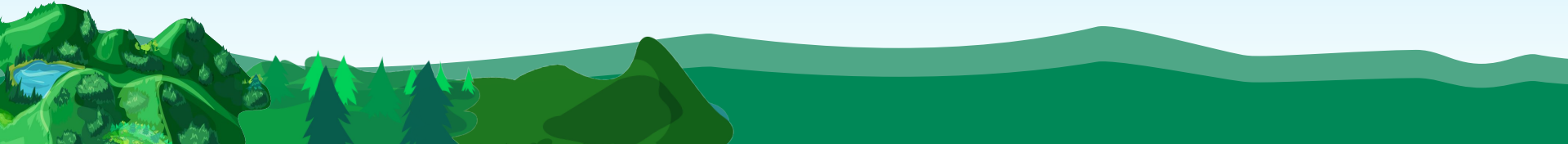
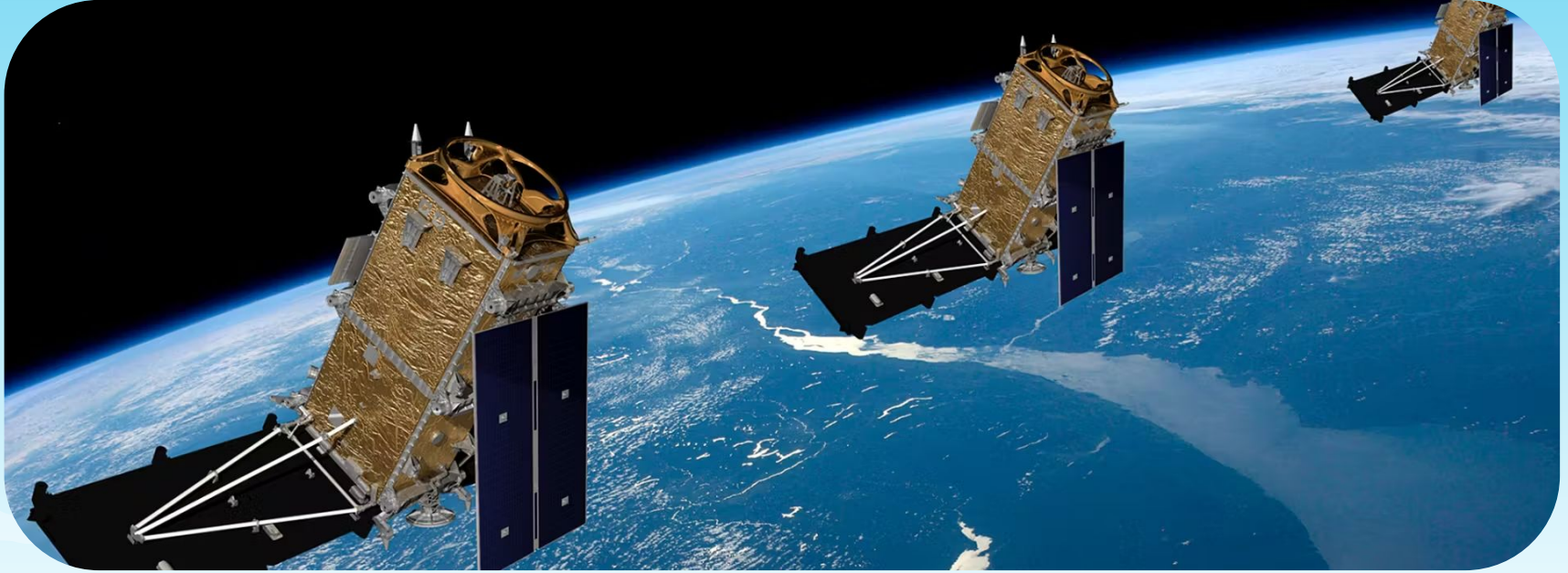
Air Traffic Disruption



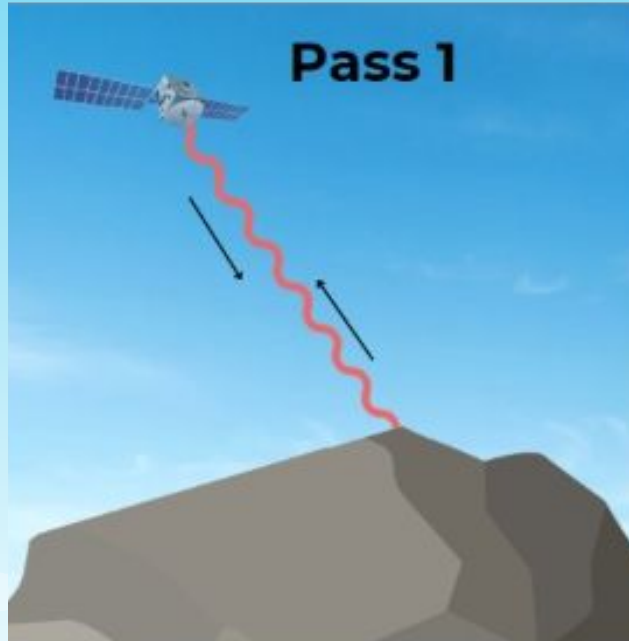
Ash Fall



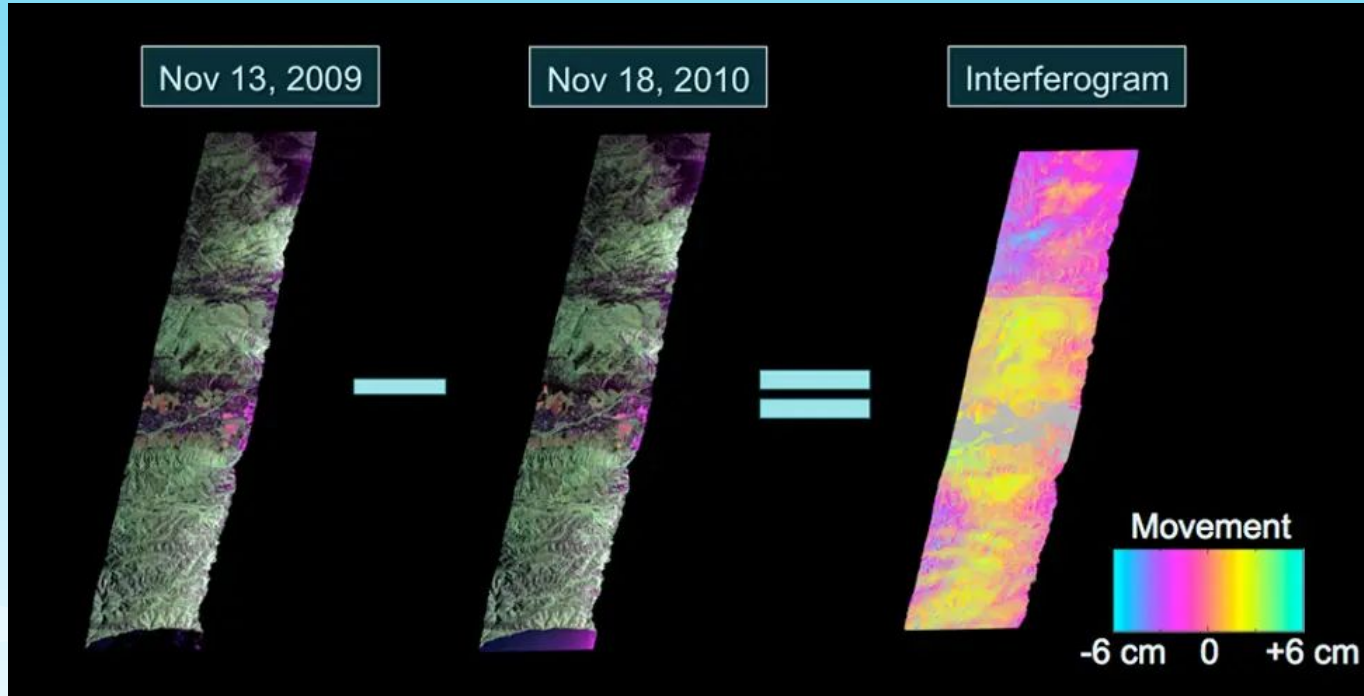
RADARSAT Constellation Mission (RCM)



Interferometric Synthetic Aperture Radar (INSAR)



Example of an Interferogram



The Problem?

Too Much Data to Analyze
Manually!



Use Machine Learning!



Project Milestones



Evaluate pre-trained CNNs on RCM InSAR data.

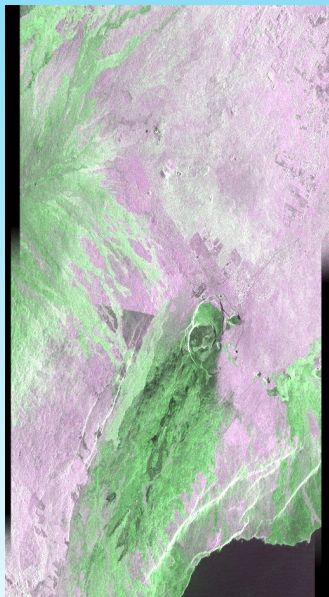


Run a sensitivity analysis to define detection thresholds.

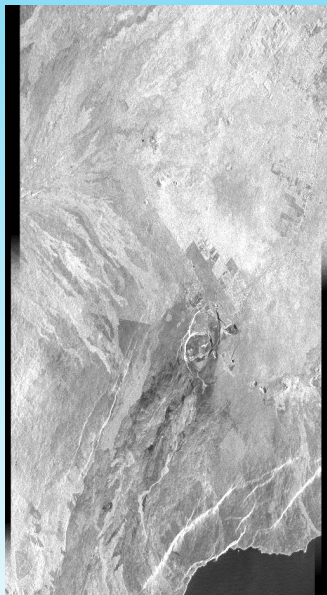


Build a scalable pipeline for near real-time detection.

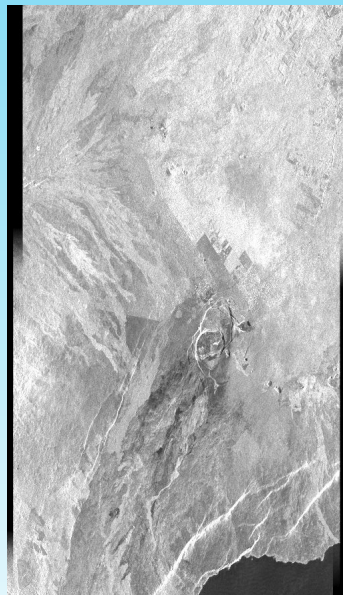
Raw SAR Images from RCM (Kilauea)



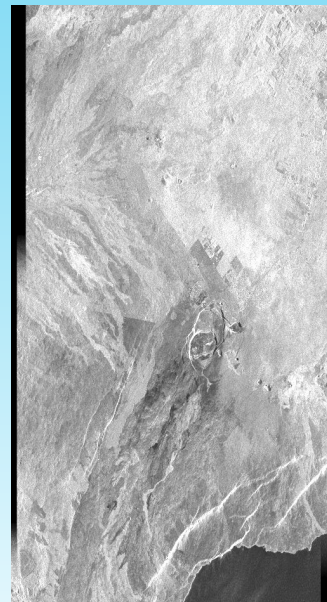
2024/05/18



2024/08/02



2024/08/10



2024/08/22

Two AI Models



Model 1

Trained on real interferograms
labels by experts

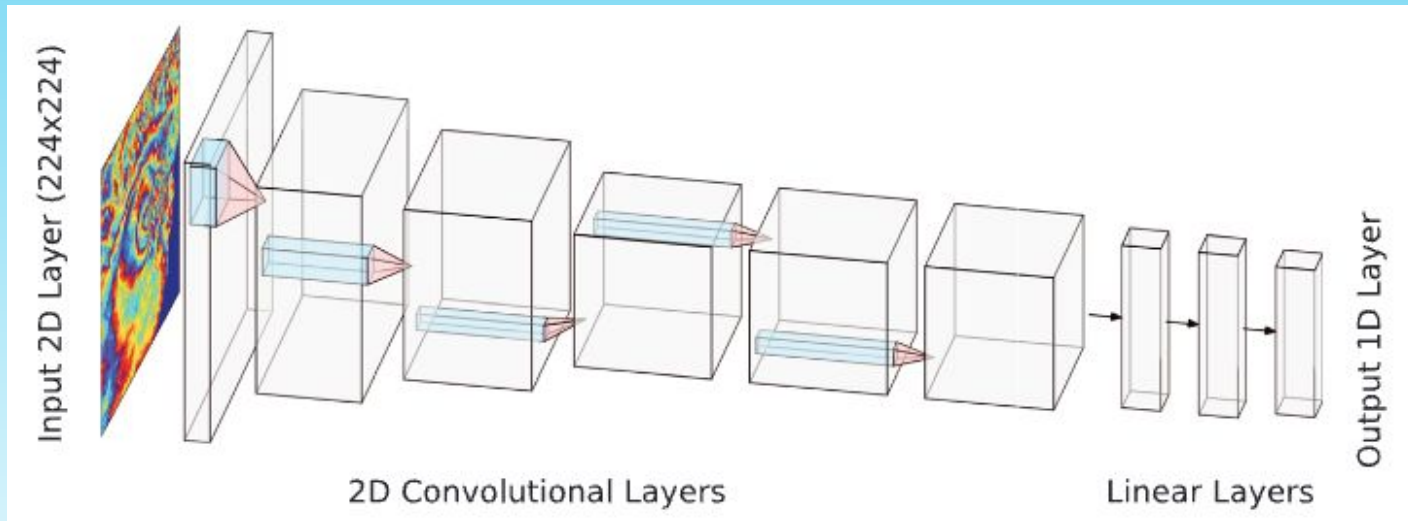


Model 2

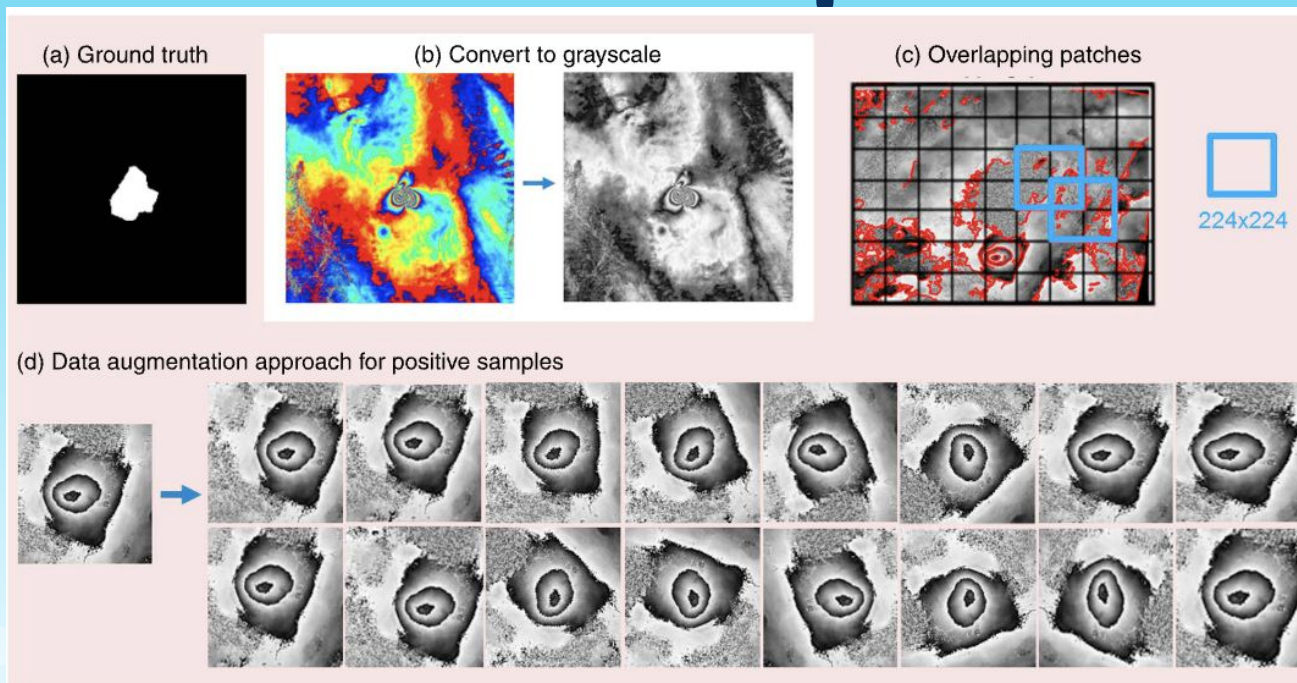
Trained on both synthetic and
real interferograms

Both Models utilize AlexNet CNN and Transfer Learning

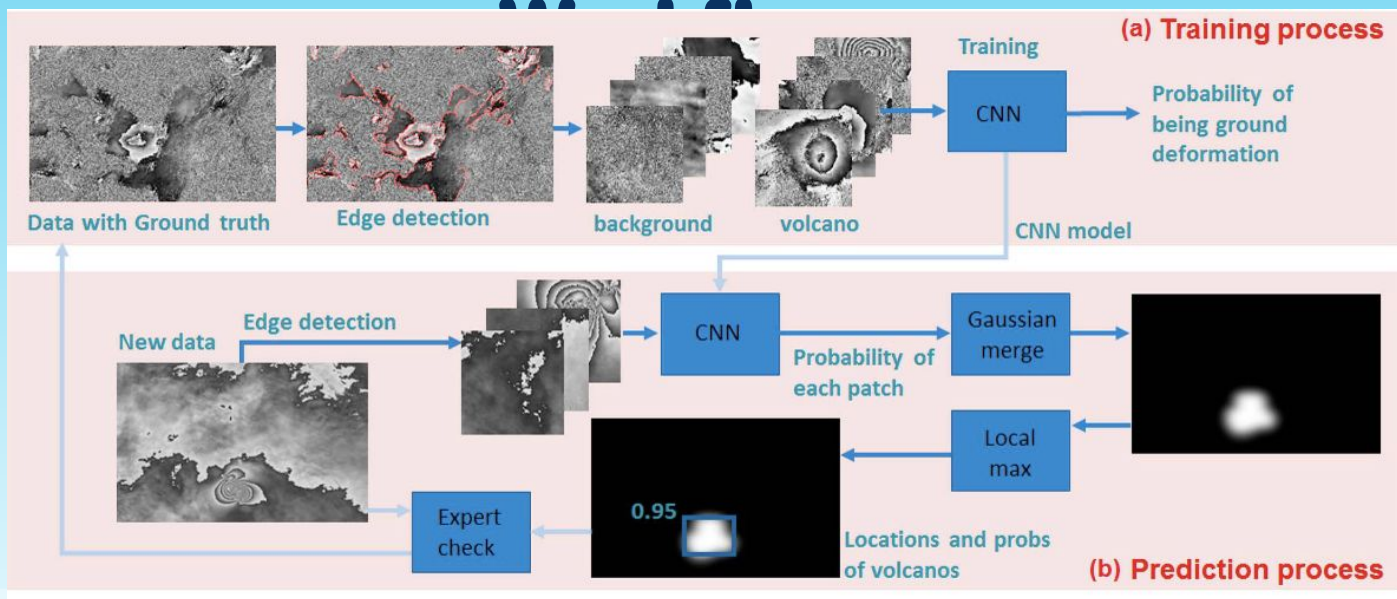
Model 1 Architecture



Model 1: Data Preparation



Model 1: Training Process & Inference



Model 1: Performance

38

hrs

Initial Training

26

hrs

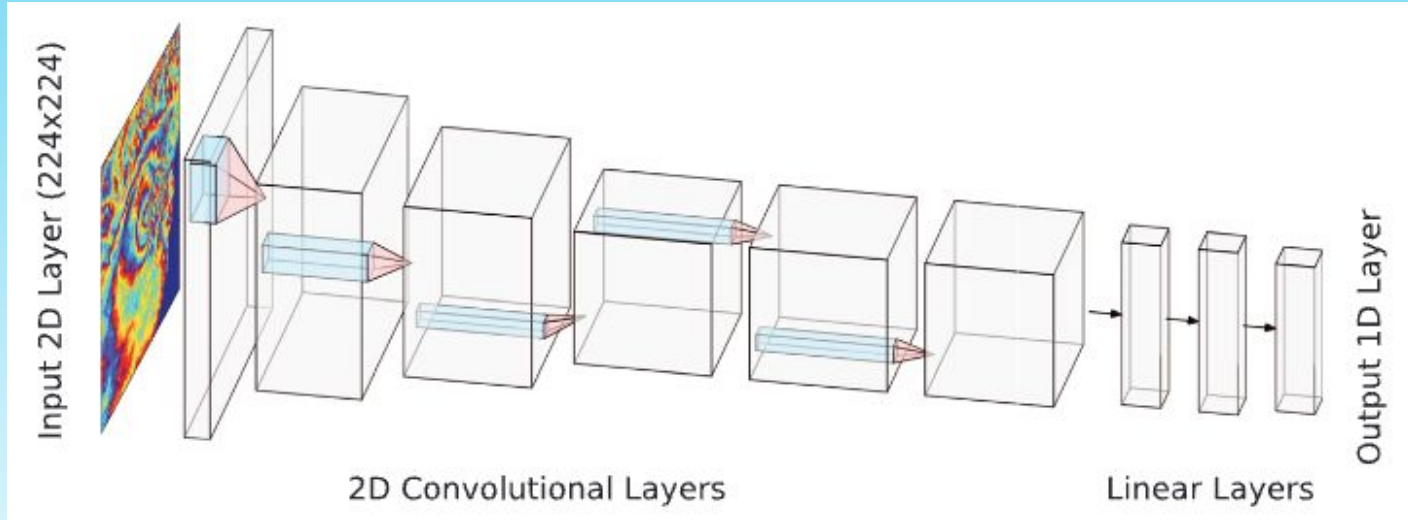
Retraining

1.5 s

Single Interferogram Inference



Model 2 Architecture



Model 2: Data Preparation

realistic synthetic interferograms by combining three components

D

Deformation

Using Mogi sources, cracks, or faults

S

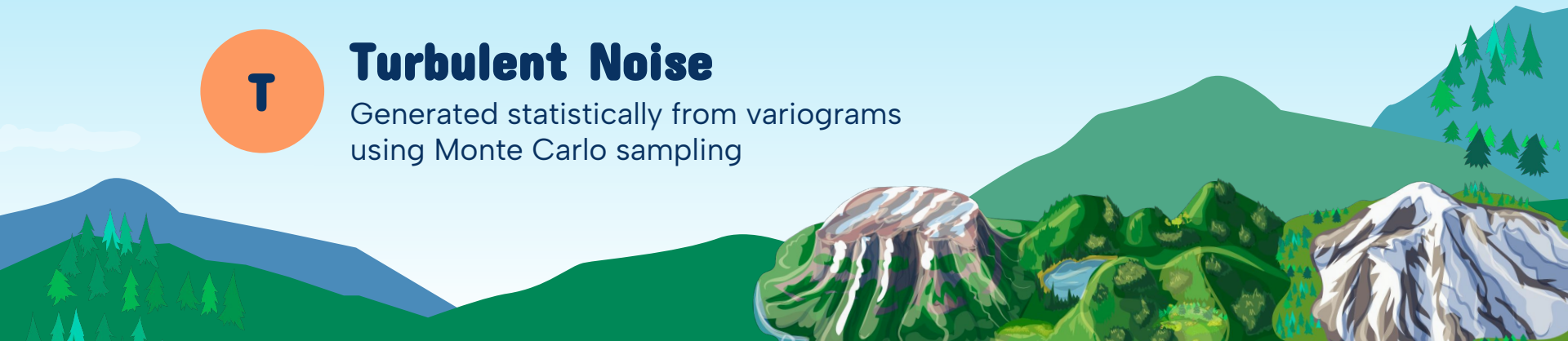
Stratified Atmospheric Delays

Using GACOS weather data

T

Turbulent Noise

Generated statistically from variograms using Monte Carlo sampling



Model 2: Three Different Variations



2-class

Deformation vs Non-Deformation



3-class

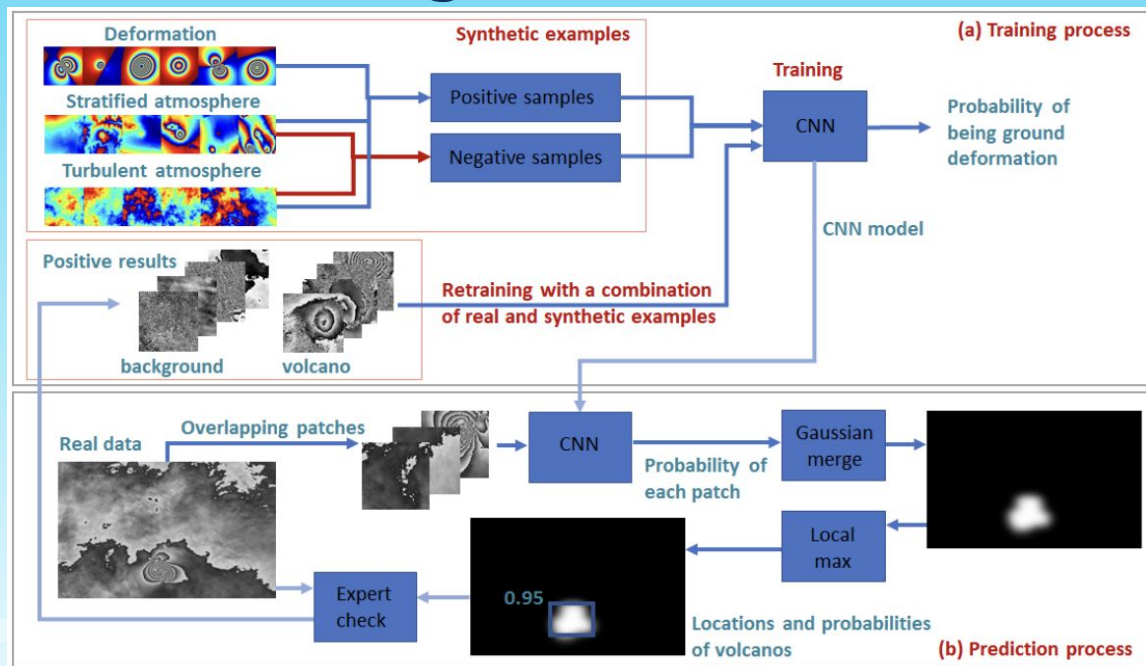
Deformation, Stratified Atmospheric Delays,
and Turbulent Noise



91-class

Predicting the weighted mix of D, S, and T as
a regression task

Model 2: Training Process & Inference



Model 2: Performance

**Up to
82%**

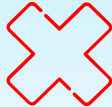
Positive Predictive Value

55%

Relative Drop in False Positives



Comparison Between Models

Feature	Model 1	Model 2
Training Data	Real	Synthetic + Real
Objective	Binary	Multi-Class & Regression
Sensitivity	Higher	Moderate
Precision	Lower	Higher
Atmospheric Correction		

Focused Volcanic Sites



Kilauea (Hawaii)

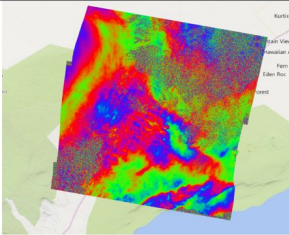
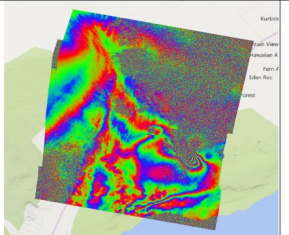
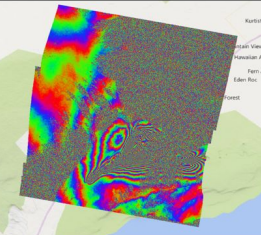


Reykjanes (Iceland)

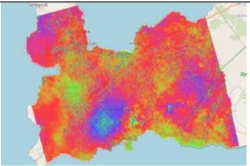
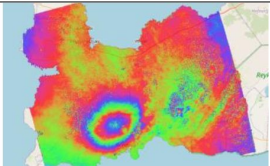
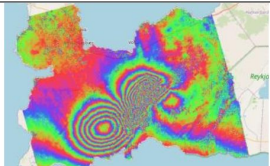


Different Deformation Levels

Kilauea (Hawaii)

Image			
Start Date	2024/08/02	2024/08/02	2024/05/18
End Date	2024/08/10	2024/08/22	2024/08/10
Deformation	Low	Medium	High

Reykjanes (Iceland)

Image			
Start Date	2024/06/29	2024/04/22	2024/07/15
End Date	2024/07/03	2024/05/04	2024/09/01
Deformation	Low	Medium	High

01

Milestone

Evaluate pre-trained CNNs on RCM InSAR data.

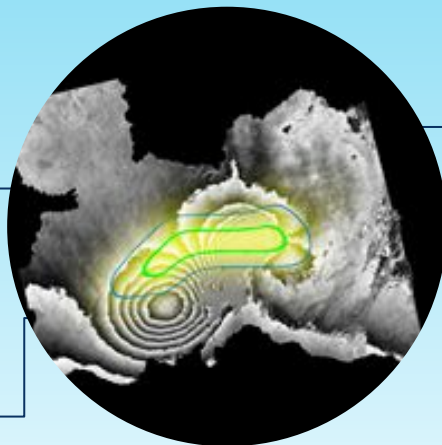


Evaluation Metrics (Milestone 1)

**Maximum Pixel
Probability**

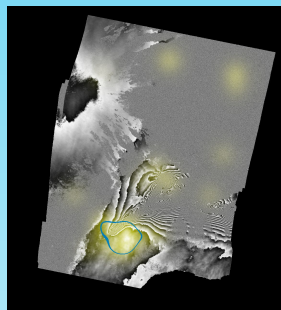
**Visual Inspection of
RGB Outputs**

**Percentage of Pixels
above 50% and 80%
Probability
Thresholds**

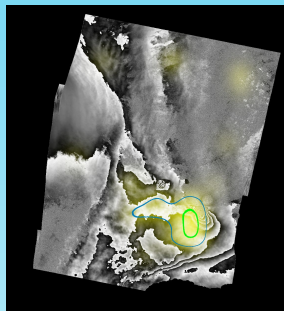


RGB Probability Map (Model 2, UTM, 45 m)

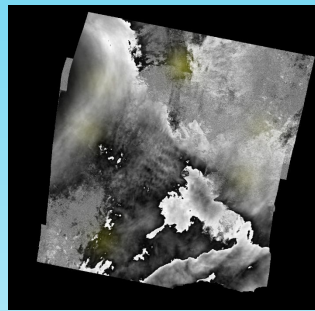
Kilauea (Hawaii)



High Deformation

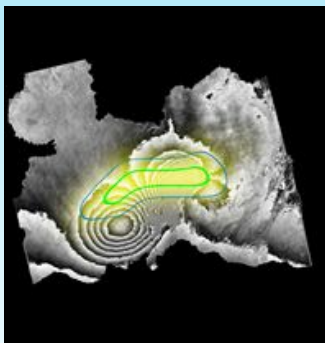


Medium Deformation

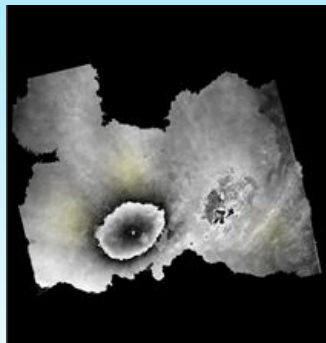


Low Deformation

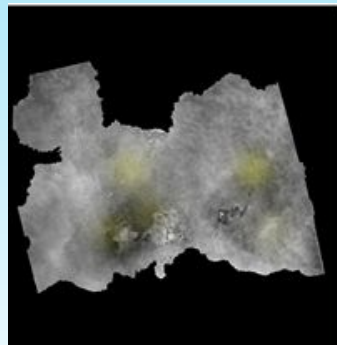
Reykjanes (Iceland)



High Deformation



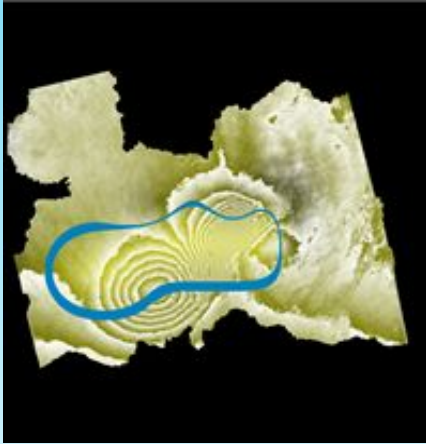
Medium Deformation



Low Deformation

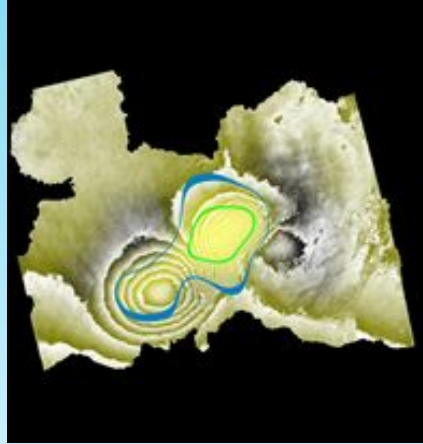
Example Model Comparison

Reykjanes High Deformation using Model 1 with UTM coordinates



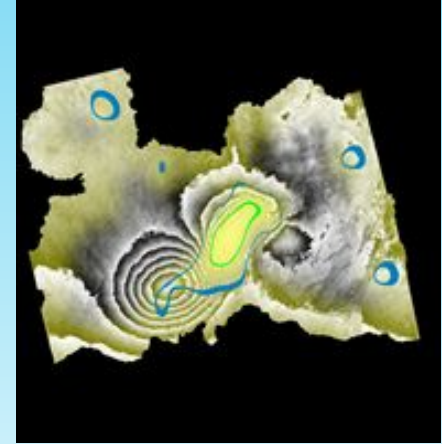
45 m

Does Not Cover the
Fringe Area Entirely



35 m

Excessive Yellowing at Edges

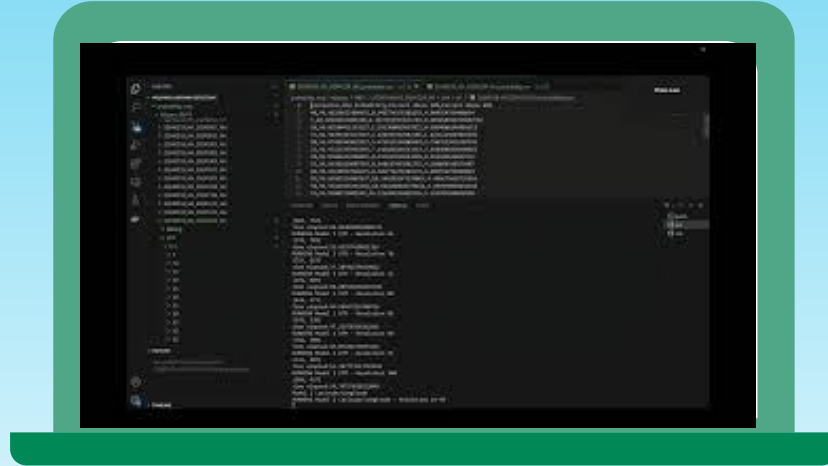


25 m

Misplaced Blue Circles



Milestone 1 Demo

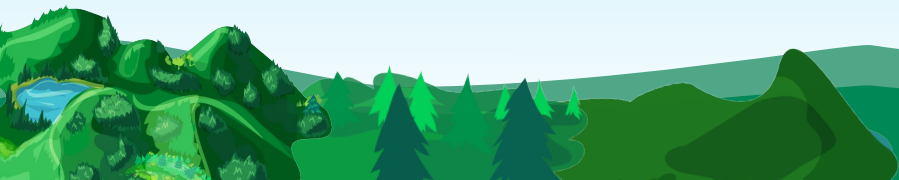


Running Model 2 on Kilauea (High Deformation)

Milestone 01:

Results

- Both Models performed better with UTM coordinates
- Model 1:
 - a. higher sensitivity,
 - b. stronger maximum probabilities in medium to high deformation events.
- Model 2:
 - a. more spatially coherent
 - b. better aligned with known deformation zones, especially in low-deformation scenarios.



02

Milestone

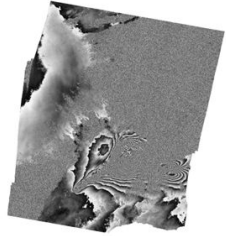
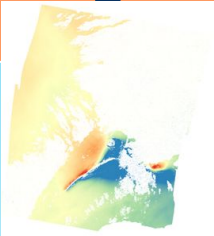
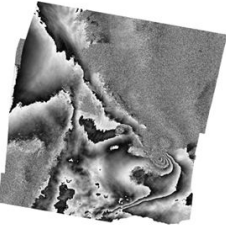
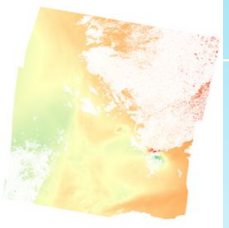
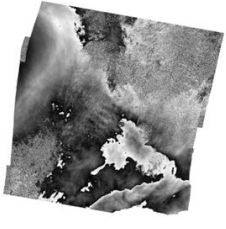
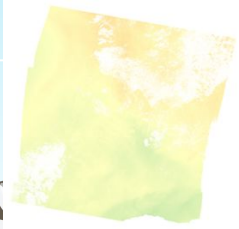
Run a sensitivity analysis to define detection thresholds.



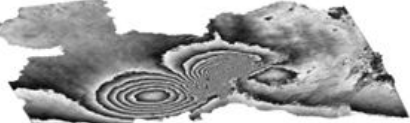
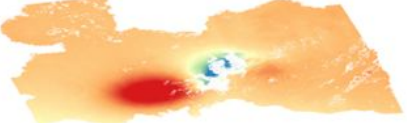
Displacement Map Legend

Value	Color	Label
-0.15		-0.1500
-0.075		-0.0750
0		0.0000
0.075		0.0750
0.15		0.1500

Kilauea(5M11)

Deformation	Date Range	Wrapped	Displacement
High	2024/05/18-2024/08/10		
Medium	2024/08/02-2024/08/22		
Low	2024/08/02-2024/08/10		

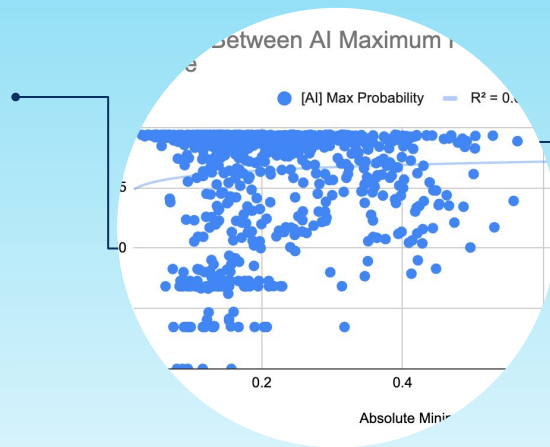
Reykjanes (5M4)

Deformation	Date Range	Wranned	Displacemen
High	2024/07/15-2024/09/01		
Medium	2024/04/22-2024/05/04		
Low	2024/06/29-2024/07/03		



Evaluation Metrics (Milestone 2)

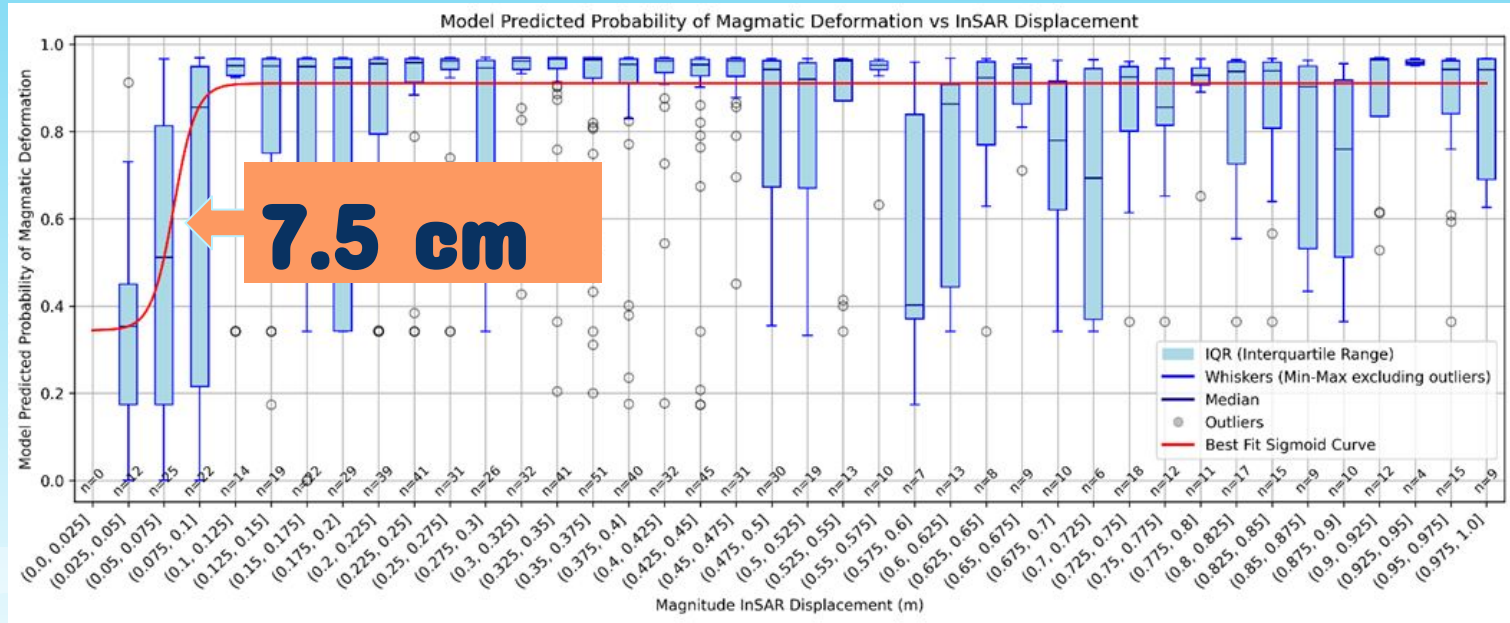
**CSV Outputs across
Time and
Deformation Levels**



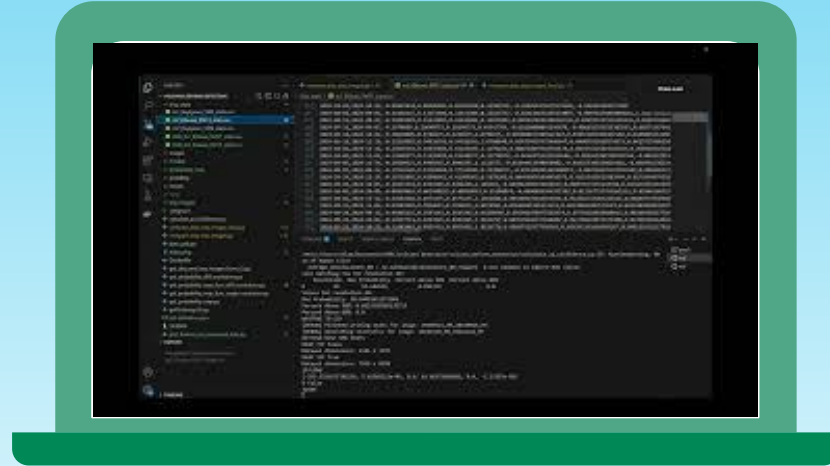
Binned Analysis

Sigmoid Curve and Box Plot

The model's predictions become more consistent and reliable at displacement levels above ~7.5 cm, while levels below that show greater variability and uncertainty.



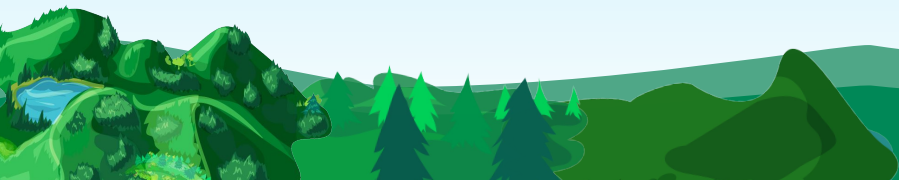
Milestone 2 Demo



Generate a CSV File for Kilauea (High Deformation)

Milestone 02: Results

- Kīlauea deformation detection threshold:
 - a. 0.03 to 0.06 normalized displacement
 - b. prediction stability was poor
- Reykjanes deformation detection threshold:
 - a. 0.08–0.22
 - b. higher accuracy and consistency
- Recall at the 80% threshold remained low for both sites
 - a. 2 to 5%



03

Milestone

Build a scalable pipeline for near real-time detection.





AWS Batch



docker

Fully containerized System Supports:

01 Pulling InSAR data from cloud storage

02 Running preprocessing and model inference

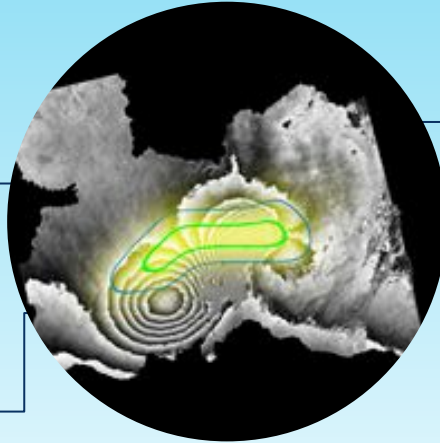
03 Outputting results for visualization and analysis



Evaluation Metrics (Milestone 3)

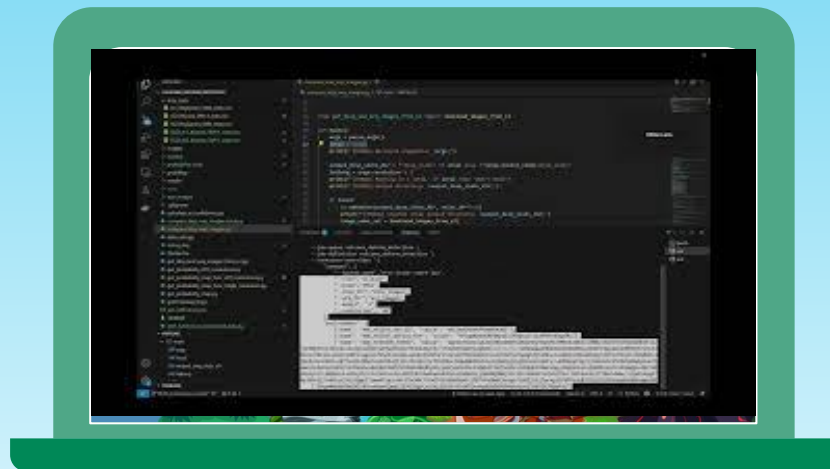
**Automation
Success Rate**

Runtime per Job



Resource Utilization

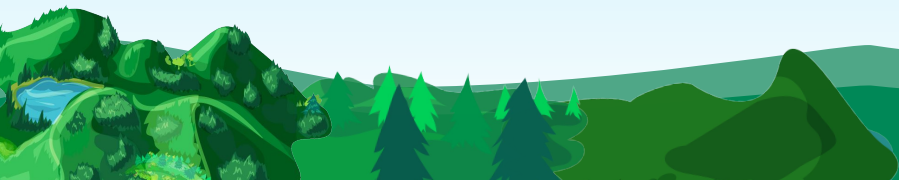
Milestone 3 Demo



Running AWS Batch Job on Kilauea (High Deformation)

Milestone 03: Results

- The system is fully containerized and supports:
 - a. Pulling InSAR data from cloud storage
 - b. Running preprocessing and model inference
 - c. Outputting results for visualization and analysis



Takeaways and Implications

Recall

[correctly identifies positive instances (true positives) from all the actual positive samples in the dataset]

Spatial Coherence

[the degree of correlation or similarity between observations at different locations]



Moving Forward



Expand the Dataset to include more Sites



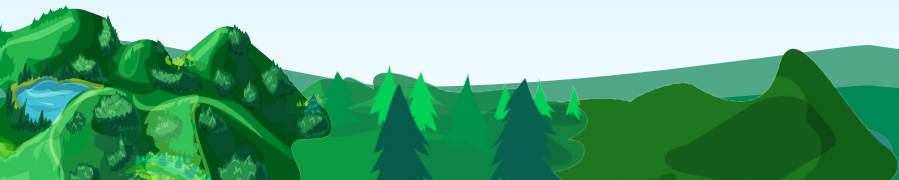
**Introduce Atmospheric Corrections
into the Pipeline**



**Improve Generalization across Diverse
Volcanic Settings**

Conclusion

- Most Spatially Coherent Outputs:
 - a. Model 2 (UTM coordinates)
- Detection performance varied by site
 - a. Higher Accuracy: Reykjanes
 - b. More Environmental Noise: Kilauea
- Sensitivity was strongest above 7.5 cm of displacement
 - a. Recall remains low—indicating a cautious model behavior.
- The system is cloud-scalable and reproducible by design



Final Thoughts





Thanks for listening!